
SC/DPS Repository Plan

A governed collection for slow-controls software, contracts, bridges, emulators, and deployment

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DUNE
DEEP UNDERGROUND
NEUTRINO EXPERIMENT

 **Fermilab**

Decision for Thursday

ASK FOR AGREEMENT

- ▶ Use an official DUNE-controlled namespace or an equivalent governed collection.
- ▶ Keep existing DUNE DAQ and firmware repositories as first-class members.
- ▶ Assign one canonical source for each schema, descriptor, and interface contract.
- ▶ Start with two vertical pilots: board-to-OPC-UA and rack-power/DPS permit.
- ▶ Exclude secrets, site endpoints, certificates, Gateway backups, and private configuration.

Recommendation

A small repository collection, not a monorepo and not untracked personal forks.

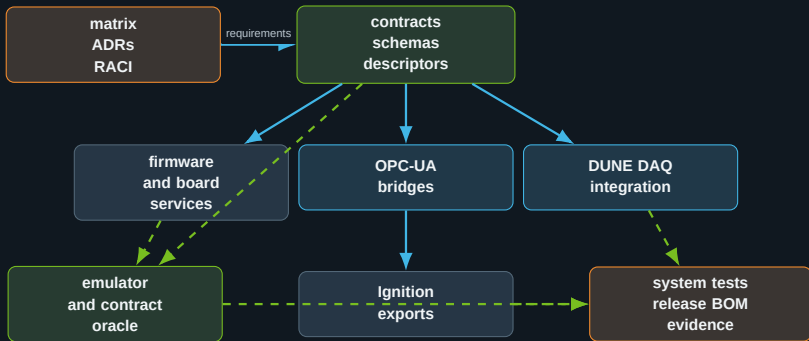
Naming

`scdps-*` for system-level repositories;
`<device>-<role>` for device implementations.

Near-term outcome

A repository catalog, owner list, branch policy, and migration checklist before wider onboarding.

Repository Relationship Topology



Lines mean implements, consumes, or verifies. The release bill of materials pins compatible versions across repositories.

Proposed Repository Collection

Repo	Owns	Seed from current work
scdps-coordination	Interaction matrix, ADRs, RACI, risks, source register, release compatibility table.	daphne_slow_controls_plan
scdps-contracts	Shared status, quality, provenance, authority, schema policy, descriptor hashes.	Extract from DAPHNE/WIB/GIZMo after interface review.
scdps-interfaces	ICD sources, intake workbooks, public review packages, controlled-source register.	sc_interface, gizmo-icd
scdps-emulator	Fleet emulator, protocol adapters, scenarios, fault injection, contract oracle.	detector-control-emulator / sc_emu
scdps-bridge-core	Reusable OPC-UA service shell, cache/quality/staleness, metrics, audit, config.	Reusable pieces from DAPHNE bridge work.
daphne-opcua-bridge, wib-opcua-bridge	Device-specific mappings, methods, tests, and source protocol pinning.	daphne-sc/opcua-bridge, wib_opcua
scdps-deployment, scdps-system-tests	Host roles, service/container definitions, release BOM, rollback, VST evidence.	Reviewed deployment files plus emulator harness.

Naming and Ownership Rules

System-level repositories

scdps-coordination

scdps-contracts

scdps-emulator

scdps-deployment

Owned by the SC/DPS collaboration team with
CODEOWNERS for specific areas

Device-level repositories

daphne-opcua-bridge

wib-opcua-bridge

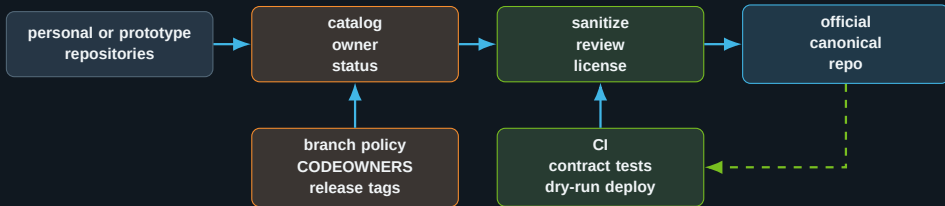
gizmo-runtime

Owned by the responsible device or subsystem team.

RULES THAT PREVENT CONFUSION

- ▶ Repo name states the ownership boundary, not the implementation language.
- ▶ Contracts are versioned artifacts, not scattered copies in every client.
- ▶ Public by default; controlled deployment values live outside Git.
- ▶ Personal forks remain development spaces, not collaboration sources of record.
- ▶ System releases pin exact compatible versions across repositories.

Onboard Existing Work Without a Restart



Initial examples: daphne-sc, daphneZMQ, wib_opcua, sc_interface;
detector-control-emulator, GIZMo public package, and gizmo-ignition.

First Vertical Pilots

Pilot 1: board to OPC-UA

DAPHNE/WIB status and readiness projected through canonical device contracts into OPC-UA, then checked against DAQ views.

read-only first

schema hash

Pilot 2: rack-power and permit

PDU, supply, sensor, and DPS-permit path exercised through emulator scenarios and one representative hardware path.

failure survives bridge loss

reset owner

Crossing fault

Inject one scenario that touches both pilots: permit loss during DAQ configure or SC command. The result should be visible in DAQ, OPC-UA/Ignition, and the evidence report, without crediting monitoring software for independent protection.

Governance Minimum for the First Repos

- ▶ Protected default and release branches.
- ▶ No direct pushes to protected branches.
- ▶ Required status checks for build, tests, and contract compatibility.
- ▶ Traceable release tags and retained artifacts.
- ▶ CODEOWNERS for schemas, firmware ABI, protection logic, deployment, and matrix rows.

Meeting output

Choose namespace, first repo names, maintainers, and migration order.

Do not decide yet

Do not force every prototype to move before the catalog and review gates exist.

Concrete next step

Create `scdps-coordination` and merge the repository catalog plus decision record.